DNS AND DHCP TROUBLESHOOTING ASSIGNMENT

Practical exercises, diagnosis, corrective actions and issue reference table

1. Simulate a DNS Resolution Failure Using the Hosts File

Objective: Demonstrate how an incorrect local hosts-file entry can override normal DNS resolution and prevent a domain name from reaching the intended website.

Procedure

1. Open Notepad as Administrator. In Windows, search for Notepad, right-click it and select Run as administrator.
2. Open the file C:\Windows\System32\drivers\etc\hosts. In the file-open dialog, change the file filter to All files so that the hosts file is visible.
3. At the bottom of the file, add an intentionally incorrect mapping. The example below uses the reserved documentation address 192.0.2.10; it is not the real Spotify server address.
4. Save the file. If Windows refuses to save it, confirm that Notepad was opened with administrator privileges.
5. Open Command Prompt and clear the local DNS cache using ipconfig /flushdns.
6. Run ping www.spotify.com and record the resolved address and the response or error. Then run nslookup www.spotify.com for comparison.

Example hosts-file entry

```
192.0.2.10    www.spotify.com
```

Expected observation and explanation

Because the hosts file is checked before querying a DNS server, Windows may attempt to ping 192.0.2.10 instead of the genuine Spotify address. The ping may show “Pinging www.spotify.com [192.0.2.10]” followed by “Request timed out” or “Destination host unreachable,” depending on routing and network conditions. The exact message can vary. The important evidence is that the domain resolves to the deliberately incorrect address.

The nslookup command normally queries the configured DNS server directly and may display the legitimate DNS answer, because nslookup does not use the Windows hosts file in the same way as normal application name resolution. This difference helps identify a local hosts-file override rather than a failure of the DNS server.

Correction and verification

7. Reopen the hosts file as Administrator and remove the incorrect line or place a # at its beginning.
8. Save the file and run ipconfig /flushdns again.
9. Run ping www.spotify.com or use ping -4 www.spotify.com. Confirm that the displayed address is no longer 192.0.2.10.
10. Test the website in a browser. A successful ping is not guaranteed because some servers block ICMP, so browser access and correct name resolution are also useful checks.

Safety note: Change only the example entry and restore the hosts file immediately after the exercise. Do not delete the other existing entries.

2. Configure DHCP with an Incorrect Default Gateway

Objective: Configure a DHCP service in Cisco Packet Tracer (or another simulator), deliberately provide an incorrect default gateway, and show how a client receives and uses that setting.

Sample topology and addressing

```
DHCP Server  ----  Switch  ----  Client PC
Network: 192.168.10.0/24
DHCP server: 192.168.10.2
Intended gateway/router LAN address: 192.168.10.1
Incorrect gateway to enter in the DHCP pool: 192.168.10.254
```

In Packet Tracer, connect the server and PC to a switch using copper straight-through cables. Configure the server with a static address such as 192.168.10.2/24. Configure the router interface connected to the switch as 192.168.10.1/24 and bring the interface up.

DHCP configuration steps

11. Open the Server device, select Services, and choose DHCP. Turn the DHCP service On.
12. Create a pool named OFFICE. Set the default gateway to the deliberately incorrect value 192.168.10.254.
13. Set the DNS server to a reachable DNS address for the lab (for example, 192.168.10.2 if DNS is configured on the server, or a suitable public DNS address when the simulator permits it).
14. Set the starting IP address to 192.168.10.100, subnet mask to 255.255.255.0, and maximum number of users to 50. Save or add the pool.
15. On the client PC, open Desktop > IP Configuration and select DHCP. Wait for the address assignment.
16. Open Command Prompt on the PC and run ipconfig /all. Record the IPv4 address, subnet mask, DHCP server and default gateway.

Expected result

```
IPv4 address:     192.168.10.100 (or another address in the pool)
Subnet mask:      255.255.255.0
Default gateway:  192.168.10.254  <-- incorrect
```

The client can communicate with devices on its local subnet, but traffic destined for other networks may fail because the configured gateway does not match the router's actual LAN address. A ping to 192.168.10.1 may succeed, while access to a remote network fails.

How to identify and fix the issue

17. Check the client's address details with ipconfig /all and note the Default Gateway field.
18. Compare the assigned gateway with the router interface address and the network plan. In this example, the correct gateway is 192.168.10.1, not 192.168.10.254.
19. On the DHCP server, edit the OFFICE pool and change the default gateway option to 192.168.10.1.
20. Renew the client's lease. On a Windows client, run ipconfig /release followed by ipconfig /renew; in Packet Tracer, switch the client to Static and back to DHCP if necessary.
21. Run ipconfig /all again and verify that the default gateway is now 192.168.10.1.
22. Test connectivity in order: ping the client's own gateway, ping a remote IP address if one exists in the simulation, and finally test access by domain name.

Conclusion: DHCP can successfully assign an IP address while still supplying an incorrect network option. Therefore, receiving a valid IP address does not by itself prove that the DHCP configuration is correct.

3. Diagnose DNS Failure When IP Connectivity Works

Scenario: Devices can reach a website or server by IP address, but cannot open it by its domain name. This usually indicates a name-resolution problem, although firewall rules and application configuration should also be considered.

Diagnostic procedure

23. Confirm basic connectivity by pinging the default gateway. If this fails, troubleshoot the local network before focusing on DNS.
24. Ping a known reachable external or lab server by IP address. If IP connectivity works but the domain name fails, DNS becomes a likely cause.
25. Run ipconfig /all and record the DNS Servers listed for the active network adapter. Check that the addresses are correct and reachable.
26. Run nslookup www.example.com. Note the DNS server used, the returned address, and any error such as DNS request timed out or Non-existent domain (NXDOMAIN). Replace the example name with the domain used in the lab.
27. Query a known working DNS server directly, where permitted, using nslookup www.example.com 8.8.8.8. Compare the result with the configured DNS server. In a closed lab, use the IP address of the lab DNS server instead.
28. Check whether the DNS service is running, whether the server has the correct forward lookup zone and records, and whether UDP/TCP port 53 traffic is permitted.
29. Correct the DNS server address or repair the DNS record/service, then clear the client cache using ipconfig /flushdns and repeat nslookup.

Useful commands and sample interpretation

```
ipconfig /all
ping 192.0.2.25
ping www.example.com
nslookup www.example.com
nslookup www.example.com 192.168.10.2
```

Command output / symptom	Likely meaning	Next check
DNS request timed out	The DNS server did not respond in time.	Check DNS server availability, IP reachability, firewall and port 53.
Non-existent domain (NXDOMAIN)	The queried name is not present in DNS or is misspelled.	Check spelling, zone name and required A/AAAA/CNAME record.
Wrong IP address returned	A record or cached answer may be incorrect.	Inspect the authoritative zone and DNS cache/TTL.
nslookup works against another server only	Configured DNS server may be misconfigured or unreachable.	Correct DHCP/manual DNS settings or repair the configured DNS server.

Solution example

If the client is configured to use 192.168.10.99 but the actual lab DNS server is 192.168.10.2, update the DHCP pool's DNS-server option to 192.168.10.2 (or correct the adapter's manual DNS setting). Renew the DHCP lease if the address came from DHCP, flush the DNS cache, and run nslookup again. Confirm that the correct IP address is returned and then test the website.

Important: A successful ping to a domain is not the only test of DNS. Some websites block ping requests. Use nslookup to directly inspect the DNS answer.

4. Troubleshoot a Client Receiving an APIPA Address

Objective: Identify why a device has self-assigned an Automatic Private IP Address (APIPA) in the 169.254.0.0/16 range instead of receiving a DHCP lease.

When a Windows client is configured for automatic addressing but cannot contact a DHCP server, it may assign itself an address in the 169.254.0.0/16 range, commonly displayed as 169.254.x.x. This normally allows only limited local-link communication and does not provide normal routed network access.

Possible causes

- DHCP server is powered off, service is stopped, or the server is unreachable.
- Client is connected to the wrong switch port, disconnected cable, disabled adapter, or faulty Wi-Fi connection.
- Switch port, VLAN, trunk, or access-port configuration places the client in the wrong broadcast domain.
- DHCP scope is exhausted and no free addresses remain in the pool.
- DHCP scope has an incorrect subnet mask, excluded range, or pool configuration.
- A DHCP relay/IP helper configuration is missing or incorrect when clients and the DHCP server are on different subnets.
- Firewall rules, access-control lists, or network security settings block DHCP traffic (UDP ports 67 and 68).
- Another network configuration issue, such as a rogue DHCP server or duplicate/conflicting settings, disrupts address assignment.

Step-by-step resolution process

30. Check the physical or wireless connection. Confirm that the network cable is connected, link lights are active, Wi-Fi is connected to the correct network, and the adapter is enabled.
31. Run ipconfig /all. Confirm that the adapter is set to obtain an IP address automatically and note whether the address begins with 169.254.
32. Run ipconfig /release and ipconfig /renew. Record any error shown when the client requests a new lease.
33. Test local connectivity to the switch/router where possible. Check the switch port status and confirm that the client is assigned to the correct VLAN.
34. Check DHCP server availability. Verify that the server is powered on, has a reachable static IP address, and that the DHCP service is running.
35. Inspect the DHCP scope. Confirm that it is active, uses the correct network and subnet mask, and has available addresses. Expand the pool or remove obsolete leases/exclusions only after checking the network plan.
36. If the server is on another subnet, verify that the router interface has the correct DHCP relay/IP helper address pointing to the DHCP server.
37. Check firewall and ACL rules to ensure DHCP requests and replies are not blocked. DHCP uses UDP port 68 on the client side and UDP port 67 on the server side.
38. After correcting the issue, renew the client lease again. Run ipconfig /all and confirm that the client receives an address from the intended subnet, the correct subnet mask, gateway and DNS server.
39. Verify connectivity by pinging the default gateway, then a reachable remote IP address, and finally a domain name using nslookup.

Troubleshooting flow

```
Client shows 169.254.x.x
        |
Check cable / Wi-Fi / adapter
        |
Check VLAN and switch port
        |
Check DHCP server and service
        |
Check scope capacity and options
```

```
        |
Check relay, firewall and UDP 67/68
        |
Renew lease -> verify valid IP, gateway and DNS
```

Expected successful result: The client receives an address from the configured DHCP pool (for example, 192.168.10.100/24), the correct default gateway (192.168.10.1), and the intended DNS server. The address should no longer be in the 169.254.0.0/16 range.

5. Common DNS and DHCP Issues: Symptoms, Causes and Solutions

The following table summarizes common problems that can occur in small office networks, home networks and simulated labs. The issues are researched examples; replace or supplement them with observations from your own practical work.

Protocol	Issue	Symptoms	Possible causes	Recommended solution
DNS	DNS server unavailable	Websites fail by name; nslookup may show a timeout.	Server is down, wrong DNS IP configured, routing or firewall blocks DNS.	Check server/service status and reachability; correct DNS settings; permit UDP/TCP 53.
DNS	Incorrect or missing DNS record	One hostname returns the wrong IP or NXDOMAIN.	A/AAAA/CNAME record is missing, misspelled or points to an old address.	Correct the zone record; verify authoritative data; flush client cache and retest.
DNS	Incorrect hosts-file entry or stale cache	One computer resolves a name differently from other devices.	Hosts file overrides DNS, or cached data is outdated.	Inspect and correct hosts file; run ipconfig /flushdns; retest with nslookup and browser.
DHCP	DHCP scope exhaustion	New clients receive 169.254.x.x or fail to obtain a lease.	All available addresses are leased or excluded.	Review active leases and pool size; reclaim obsolete leases or expand the scope safely.
DHCP	Incorrect default gateway option	Client gets an IP address but cannot reach other networks.	DHCP option 003 supplies the wrong router address.	Correct the gateway in the DHCP scope; renew the client lease; verify with ipconfig /all.
DHCP	DHCP server or relay unavailable	Clients cannot obtain an address, especially across subnets.	DHCP service is stopped, server is offline, or relay/IP helper is missing.	Restore DHCP service/server connectivity; configure and test the correct relay address.
DHCP	Wrong VLAN or blocked DHCP traffic	Some clients fail while others on a different segment work.	Switch port VLAN is incorrect or ACL/firewall blocks UDP 67/68.	Correct VLAN/port settings and permit DHCP request/reply traffic.

Overall conclusion

DNS and DHCP perform different but closely related functions. DHCP automatically provides clients with IP configuration, including the default gateway and DNS server. DNS translates hostnames into IP addresses. A mistake in either service can make network access appear unavailable. A reliable troubleshooting method is to check the physical connection first, verify the client's IP configuration, test gateway and IP connectivity, inspect DHCP lease information, and then use nslookup to diagnose name resolution.

Practical evidence to capture

- Screenshot of the hosts-file test showing the intentionally incorrect mapping (hide unrelated personal information).
- Command Prompt output showing ping www.spotify.com resolving to the test address, followed by the corrected result.
- Packet Tracer topology and client IP Configuration or ipconfig /all output showing the incorrect gateway, then the corrected gateway.
- nslookup output showing the DNS failure and the successful result after correction.
- Client ipconfig /all output showing the APIPA address before troubleshooting and a valid DHCP lease after correction.

Note: The sample addresses and command outputs in this assignment are illustrative. Your actual ping errors, assigned addresses, simulator screens and nslookup results may differ. Capture screenshots from your own computer or simulator for genuine practical evidence.